Ayomide Abilewa

Electronic and Electrical Engineering student, focused on embedded systems and instrumentation

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RESEARCH PROFILE

Electronic and electrical engineering student at Obafemi Awolowo University, expected 2027, working on measurement and detection under non-ideal conditions. Recent work covers a two-stage cascaded detector that fuses RGB and edge-enhanced models for pipe anomaly identification, a multi-sensor multispectral acquisition node, and control-systems practice on Quanser robotics hardware using MATLAB and Simulink.

EDUCATION

B.Sc. Electronic and Electrical Engineering · Obafemi Awolowo University — 2021–2027 (expected)

Ile-Ife, Osun State, Nigeria

Relevant coursework: Control Systems, Measurement and Instrumentation, Digital Electronics, Signal Processing, Power Systems, Microprocessors and Embedded Systems, Electronic Circuit Design.

RESEARCH INTERESTS

- Detection under degraded conditions. How to keep detection precision steady when lighting, surface condition and viewing angle vary within a single run, instead of tuning for one condition and losing the others.
- Measurement and calibration. Turning raw sensor output into a measurement that means something: calibration, separating quantities that are easy to conflate, and the signal standards industry already relies on.
- Feedback and control. Closed-loop behaviour on real hardware, from a thermistor-driven PWM loop to PID control on Quanser platforms, where the actuator changes the quantity being measured.
- Embedded and real-time vision. Running vision pipelines within real constraints: modest hardware, live video, and the requirement that the system respond now rather than after a batch job.
- Systems that degrade rather than fail. Designing explicit fallback behaviour so losing a network, a model or a sensor costs quality instead of function.

RESEARCH AND TECHNICAL PROJECTS

Live Video Pipe Anomaly Detection · Python, YOLOv8, YOLOv11, Weighted Boxes Fusion — 2026 – present

- Developing a two-stage cascaded detection architecture for real-time pipe anomaly identification, targeting industrial inspection.
- Built a parallel ensemble of RGB and edge-enhanced YOLO models merged with Weighted Boxes Fusion to improve precision across varied lighting and surface conditions.

Multispectral Acquisition Device for Hydroponics · ESP32, GY-2561, GY-31, I2C — 2025

- Built a multi-sensor ESP32 acquisition node measuring light intensity and colour spectrum with the GY-2561 and GY-31, streaming live to a Blynk IoT Cloud dashboard.
- Integrated multiple I2C sensors on a single ESP32, handling bus arbitration and sensor calibration.

AI-Powered Smart Attendance System · Python, OpenCV, YuNet, SFace — Feb – Mar 2026

- Built an AI-powered attendance system using YuNet face detection and SFace recognition with cosine similarity matching, delivering the computer-vision pipeline and a Flask REST API backend.
- Evaluated recognition performance under real classroom lighting and off-axis faces rather than curated frontal images.
- Defended the system at a final-year group project review and was invited to develop an extended production version.

github.com/ayomide-abilewa/Smart-Attendance-System

Smart Temperature-Controlled Fan · Arduino, C++, PWM, NTC thermistor — 2024

- Implemented continuous proportional actuation rather than threshold switching to avoid overshoot.

RESEARCH AND ENGINEERING EXPERIENCE

Research and Development Intern (SIWES) · ACE Quanser Robotics Lab, Obafemi Awolowo University 2024

Ile-Ife, Nigeria

- Operated the QDrone quadrotor UAV, QCar autonomous ground vehicle and QArm 6-DOF manipulator using MATLAB and Simulink.
- Applied PID control, sensor feedback and real-time system analysis to live hardware rather than simulation alone.
- Delivered a technical orientation presentation on the Quanser Aero 2 covering system architecture, control principles and laboratory safety.

Facilities Engineering Management Intern (SIWES) · Chevron Nigeria Limited Mar 2025–Present

Lagos, Nigeria

- Rotate across the Electrical and Instrumentation (E&I) team, working on oil and gas field instrumentation and electrical systems.
- Interpret P&IDs and study control loop configurations, pneumatic signal standards (3–15 PSI), I/P converters and hazardous area classification.

Embedded Systems and EEE Tutor · Astar Tutorials 2023–Present

Ile-Ife, Nigeria

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, JavaScript, Verilog/HDL, Bash

Embedded & Hardware: Arduino, ESP32, STM32, Raspberry Pi, FPGA, I2C / SPI / UART, Sensor integration

AI & Machine Learning: YOLOv8/v11, OpenCV, TensorFlow (fundamentals), PyTorch (fundamentals), Weighted Boxes Fusion

Instrumentation & Control: P&ID interpretation, PID control loops, Pneumatic systems, I/P converters, 4–20 mA and 3–15 PSI standards

Tools: VS Code, Git/GitHub, MATLAB/Simulink, KiCad, Flask, React/TypeScript, PostgreSQL, Blynk IoT

TRAINING AND CERTIFICATIONS

- FPGA Design and HDL/Verilog Workshop (SWEP). Faculty of Engineering, Obafemi Awolowo University, 2024.
- Robotics Lab Orientation and Operations. ACE Quanser Lab, Obafemi Awolowo University, 2024.

TEACHING AND ACADEMIC SERVICE

SPAW Project Lead and Syllabus Team Lead · IEEE OAU Student Branch 2024–Present

- Lead SPAW 3.0, a six-session embedded systems and entrepreneurship curriculum for secondary school students.